International Boundary and Water Commission, US Section

LAN HVAC System Replacement

Scope of Work

Project Title:	HVAC System Replacement
Agency/Department:	International Boundary and Water Commission, US Section (USIBWC) / Information Management Department (IMD)
Location:	Headquarters LAN Room - 4191 N. Mesa St, El Paso, TX 79902

1. Purpose

The purpose of this Scope of Work is to define the requirements for the removal of the existing air-cooled Leibert HVAC system and installation of a new system of similar capacity and functionality in the headquarters LAN room, including integration with the existing Carrier HVAC system via a damper for airflow control. The Contractor shall furnish all labor, materials, equipment, and services necessary to complete the work in accordance with applicable federal, state, and local regulations, including the Federal Acquisition Regulation (FAR).

2. Background

The LAN room houses critical networking equipment requiring precise temperature control. The existing HVAC system utilizes R-22 refrigerant and requires replacement to meet current environmental and operational standards. The new system shall employ R-410A refrigerant or an alternative refrigerant as prescribed by current regulatory requirements and shall conform to all applicable codes and standards. The existing Carrier HVAC system will remain in place and must be integrated with the new system using a motorized damper to switch cooling air between units.

3. Requirements

The Contractor shall perform the following tasks:

3.1 Equipment

- Provide and install one (1) PX023UA1A809KM Liebert PDX Air-Cooled System with remote condensing unit or equivalent system meeting or exceeding performance specifications.

3.2 Damper Installation

- Provide and install a motorized damper system to allow switching of cooling air between the new Liebert unit and the existing Carrier HVAC system.
- Damper must be integrated with the HVAC control system for automated or manual operation.

3.3 Removal and Installation

- Provide crane service to remove existing equipment and install new equipment.
- Remove and dispose of old equipment in compliance with EPA regulations and FAR Part 23 – Environment, Energy and Water Efficiency, Renewable Energy Technologies, Occupational Safety.

3.4 Refrigeration Piping

- Provide and install all required refrigeration piping.
- Ensure compatibility with the selected refrigerant.

3.5 Electrical

- Reconnect new equipment to existing electrical infrastructure.

3.6 Startup and Commissioning

- Perform full startup of new equipment.
- Log and provide operating temperatures and pressures.

4. Performance Specifications

The proposed system must meet or exceed the following minimum specifications:

- **Cooling Capacity:** 23 kw (or equivalent to PX023UA1A809KM model).
- **Refrigerant Type:** R-410A or any other refrigerant recommended or required to comply with current regulations.
- **Energy Efficiency:** Minimum EER of 10.0 or higher.
- **Voltage:** 460V, 3-phase, 60Hz.
- **Operating Temperature Range:** 65°F to 95°F ambient.
- **Controls:** Digital microprocessor-based control system with remote monitoring capability.
- **Damper Control:** Motorized damper integrated with HVAC control system.
- **Compliance:** Must meet ASHRAE standards and applicable federal energy efficiency requirements.

5. Warranty Requirements

- **Minimum Warranty:**
    - Equipment: 1-year parts and labor; 5-year compressor warranty.
- Vendors are encouraged to include extended warranty options in their proposal.

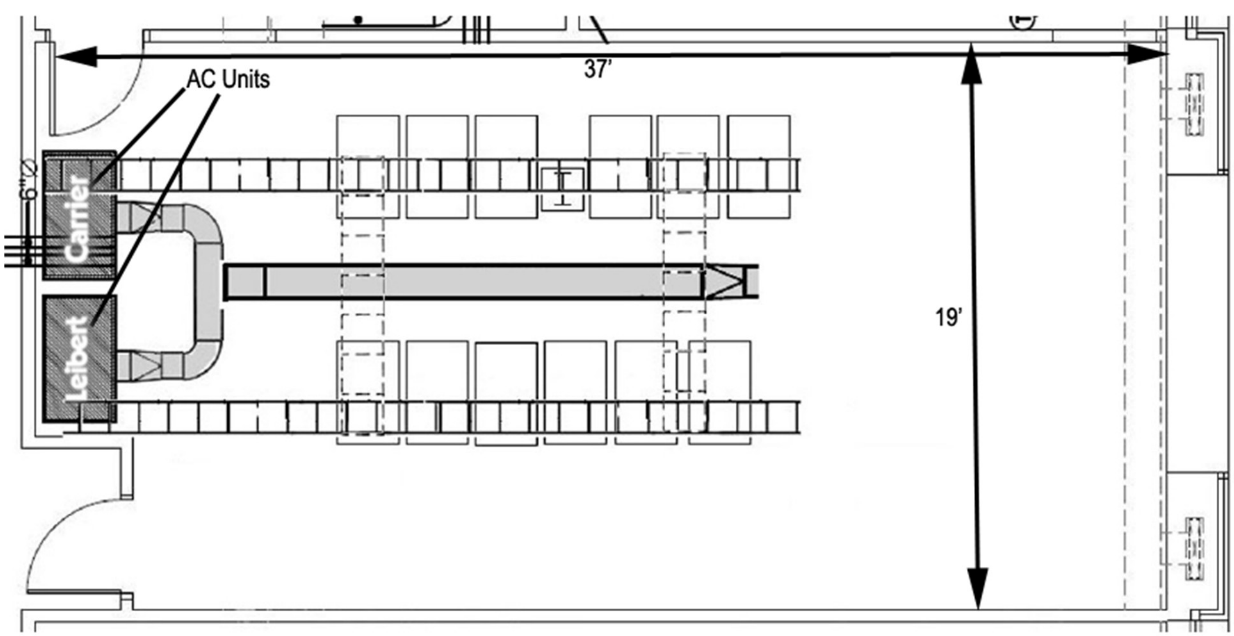
6. Deliverables

- Installed and operational HVAC system and damper in Headquarters LAN Room.
- Startup report with operating temperatures and pressures.
- Disposal documentation for removed equipment.
- Complete Operation & Maintenance (O&M) manuals for all installed equipment and components, provided in both hard copy and digital format.

8. Proposal Submission Requirements

- Technical proposal including equipment specifications.
- Cost proposal with detailed breakdown.
- Project schedule.
- Warranty information (minimum and extended options).

Diagram of LAN Room:



Leibert AC unit width: 32.5 in.

Leibert AC unit height: 76.5 in.

Height from floor to duct: 96 in.

Distance from wall to Leibert AC unit: 11 in.

Distance from current Leibert AC unit to Carrier AC unit: 32.5 in.

Picture shows measurement from wall to LAN Room entrance.

Distance from wall nearest Leibert to LAN room entrance: 154.25 in. or 12 ft, 10.25 in.